

Advanced programming Assignment

ROLL NO: CSB24065

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Question:

Assignment 14: Create a scenario where objects are "dead" but still have a reference count higher than zero, then force the Garbage Collector to clean them up. Do in python only.

Implementation Steps:

Create a Node class with a name and a link attribute.

Create a Cycle: Instantiate Node A and Node B.

Set A.link = B and B.link = A.

Check References: Use `sys.getrefcount()` to show that both objects have multiple references.

The "Deletion": Use `del A` and `del B`.

The Investigation: Use the `gc` module to show that these objects still exist in memory because of the cycle, even though you can no longer access them from your code.

The Cleanup: Call `gc.collect()` and print the number of "unreachable" objects collected.

Answer:

```
import sys
```

```
import gc
```

```
class Node:
```

```
    def __init__(self, name):
```

```
        self.name = name
```

```
        self.link = None
```

```
    def __del__(self):
```

```
        print(f"{self.name} is being destroyed")
```

Enable garbage collector

```
gc.enable()
```

```
print("Creating nodes...\n")
```

```
A = Node("Node A")
```

```
B = Node("Node B")
```

Creating circular reference

```
A.link = B
```

```
B.link = A
```

```
print("Cycle created:")
```

```
print("A ->", A.link.name)
```

```
print("B ->", B.link.name)
```

Checking reference counts

```
print("\nReference Counts:")
```

```
print("A:", sys.getrefcount(A))
```

```
print("B:", sys.getrefcount(B))
```

Save object ids for investigation

```
a_id = id(A)
```

```
b_id = id(B)
```

```
print("\nDeleting A and B variables...\n")
```

```
del A
```

```
del B
```

Objects still exist because of circular reference

```
print("Objects are still in memory because of the cycle.")
```

```
found = False
```

```
for obj in gc.get_objects():
```

```
    if id(obj) == a_id or id(obj) == b_id:
```

```
    print("Found unreachable object still in memory:", obj.name)
    found = True

if not found:
    print("Objects not found.")

print("\nRunning Garbage Collector...\n")

collected = gc.collect()

print("Unreachable objects collected:", collected)

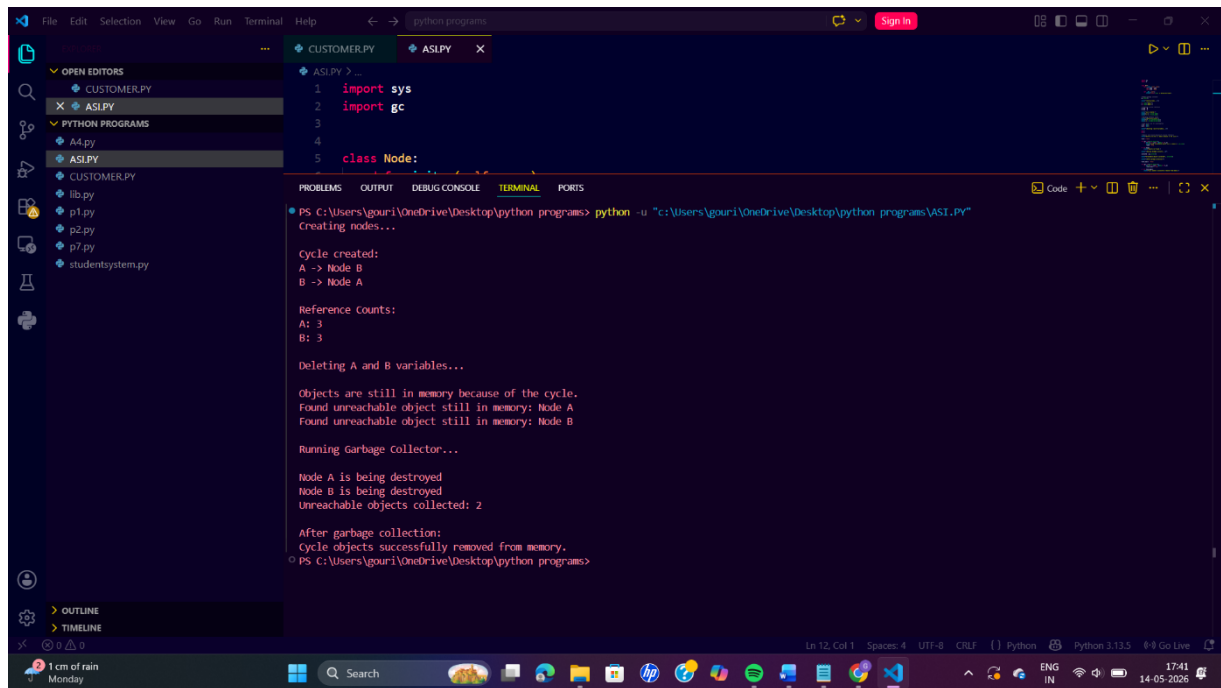
print("\nAfter garbage collection:")

found_again = False

for obj in gc.get_objects():
    if id(obj) == a_id or id(obj) == b_id:
        found_again = True

if not found_again:
    print("Cycle objects successfully removed from memory.")
```

Screenshots of the output:



```
File Edit Selection View Go Run Terminal Help python programs
CUSTOMER.PY ASLPY
CUSTOMER.PY
ASLPY
PYTHON PROGRAMS
A4.py
CUSTOMER.PY
lib.py
p1.py
p2.py
p7.py
studentsystem.py
OUTLINE
TIMELINE
1 cm of rain
Monday
17:41
14-05-2026
```

```
1 import sys
2 import gc
3
4
5 class Node:
```

```
PS C:\Users\gauri\OneDrive\Desktop\python programs> python -u "c:\Users\gauri\OneDrive\Desktop\python programs\AST.Py"
Creating nodes...

Cycle created:
A -> Node B
B -> Node A

Reference Counts:
A: 3
B: 3

Deleting A and B variables...

Objects are still in memory because of the cycle.
Found unreachable object still in memory: Node A
Found unreachable object still in memory: Node B

Running Garbage collector...

Node A is being destroyed
Node B is being destroyed
Unreachable objects collected: 2

After garbage collection:
Cycle objects successfully removed from memory.
PS C:\Users\gauri\OneDrive\Desktop\python programs>
```